

# KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

## MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

**Subject: Data Structures**

**Time: 1.5 Hours**

**Semester / Programme: 3rd Semester**

**Full Marks: 20**

**Instructions:** Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

### SECTION - A (Compulsory)

**1(a)** Calculate the total number of different Binary Search Trees (BST) that can be constructed using  $n = 5$  distinct keys. [1]

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

**1(b)** Convert the following infix expression into its equivalent postfix notation:  $A + B \times C - (D / E F)$ . [1]

**1(c)** State the maximum number of nodes at level  $l$  (assuming root is at level 0) in a Binary Tree. [1]

$$N_l = 2^l$$

**1(d)** Evaluate the address of element  $A[15][10]$  in a 2D integer array  $A[1020][515]$  stored in Column-Major order, given base address  $\{Base\}(A) = 1000$  and each integer occupies 4 bytes. [1]

$$\text{Address}(A[i][j]) = \text{Base} + W \times ((i - LB_r) + (j - LB_c) \times M)$$

**1(e)** What is the primary advantage of a Circular Queue over a Linear Queue implemented using arrays? [1]

### SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

**Q.2. (a)** Write an efficient algorithm/C function to represent a sparse matrix using a 3-tuple (triple) linked list or array representation, and explain its storage advantage. [2.5]

**(b)** Convert the infix expression  $Q = ((A + B) \times C - (D - E)) \div (F + G)$  into prefix notation using stack-based evaluation or operator precedence method. [2.5]

**Q.3. (a)** Write a complete C function to reverse a singly linked list in-place. Show the pointer manipulations clearly. [3]

**(b)** Explain how two polynomials in a single variable can be represented using linked lists. Write the structural node definition in C. [2]

**Q.4. (a)** Construct a Binary Tree given its Preorder traversal: `G, B, A, C, F, E, D, H, I` and Inorder traversal: `A, B, C, D, E, F, G, H, I`. [3]

**(b)** Write a recursive C function to compute the height (or maximum depth) of a given Binary Tree.  $\text{Height}(T) = 1 + \max(\text{Height}(T_{\text{left}}), \text{Height}(T_{\text{right}}))$  [2]

**Q.5. (a)** Starting with an empty Binary Search Tree (BST), insert the following keys in the given order: `50, 30, 70, 20, 40, 60, 80`. Draw the resulting BST and perform its Post-order traversal. [3]

**(b)** Write a non-recursive (iterative) algorithm to traverse a Binary Tree in In-order using an explicit stack. [2]

**--- END OF PAPER ---**